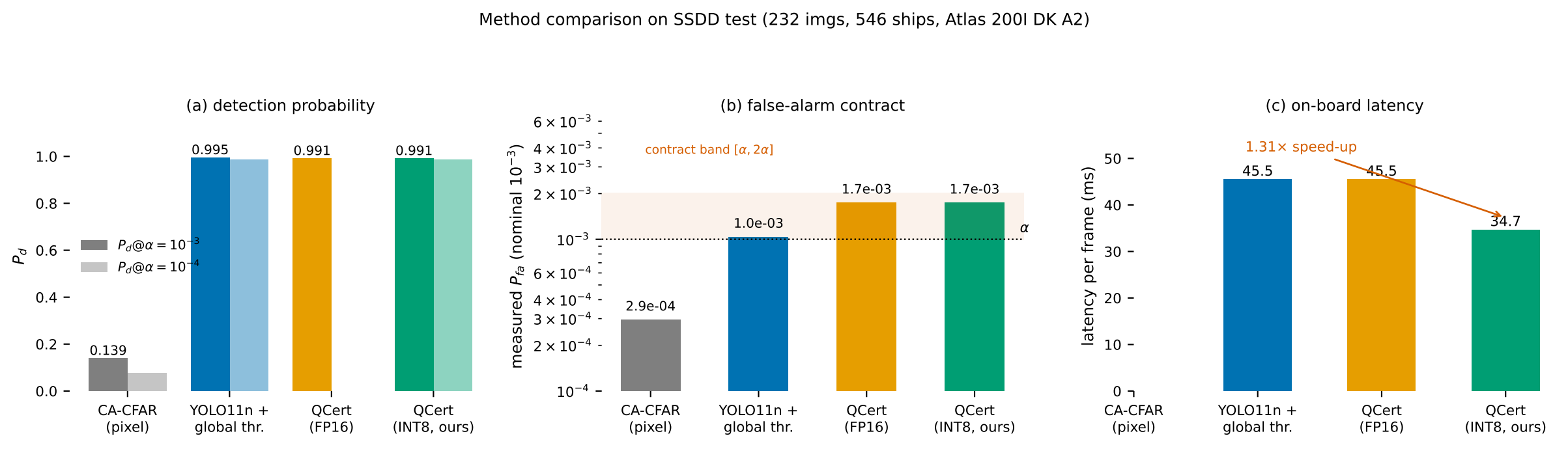
Method comparison on SSDD test (232 imgs, 546 ships, Atlas 200I DK A2)
(a) detection probability
P_d
$P_d@\alpha = 10^{-3}$
$P_d@\alpha = 10^{-4}$
0.139
0.995
0.991
0.991
CA-CFAR (pixel)
YOLO11n + global thr.
QCert (FP16)
QCert (INT8, ours)
(b) false-alarm contract
measured P_{fa} (nominal 10^{-3})
contract band $[\alpha, 2\alpha]$
α
2.9e-04
1.0e-03
1.7e-03
1.7e-03
(c) on-board latency
latency per frame (ms)
1.31× speed-up
45.5
45.5
34.7